

Education and Certifications

- **M.Sc. Computer Science**, Institute of Computing - UNICAMP, Brazil. **2023–07/2026**
- **Teste Internacional TOEFL iBT**, 99/120, C1 level - advanced **2023**
- **B.Sc. Mechanical and Electrical Engineering**, Universidad Nacional de Ingeniería - UNI, Peru. **2011–2016**
- I completed several courses on ML, TensorFlow, TensorFlow JS, TensorFlow Lite, Big Data and Analytics with AWS, TinyML, IoT, Smart Cities, Web Development, Linux, React.js, etc. [Link](#).

Professional Experience

<u>Software Developer and Instructor</u>	Aubo	2019 - 2023
IoT + Cloud + AI Solutions	Lima, Perú	

- A small company I founded to create solutions using IoT + Cloud + AI, as well as training programs.

<u>IoT Instructor</u>	Big Data IoT Cloud	2021
Course on IoT, AI, Cloud and Blockchain	Lima, Perú	

- IoT workshops using cloud platforms AWS, GCP, and Azure.
- IoT communication technologies used: LoRaWAN, Sigfox and WiFi.
- Implementation of edge solutions (TinyML) with TensorFlow Lite and Edge Impulse.
- Implementation of IOTA (blockchain designed for IoT) on an AWS virtual machine.

<u>IoT Researcher</u>	Research Institute of the Faculty of Electronics - UNI	2019–2021
	Lima, Perú	

- Principal researcher on 3 research projects.
- IoT research on Smart Campus published in IEEE and Springer.

<u>Co-founder</u>	IoT Community Peru	2018–Present
Non-profit community for spreading IoT awareness	Lima, Perú	

- More than 600 members on WhatsApp.
- More than 4000 followers on LinkedIn.
- Nearly 700 subscribers on YouTube with 50 videos about IoT.

<u>IoT Researcher</u>	IoT Research Group - UNMSM	2018 - 2023
	Lima, Perú	

- IoT and ML research, award at a technology fair.

Research Papers

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- **Proposal and Evaluation of a Software Architecture for a Smart Campus Digital Twin**: master's thesis project where I built an IoT system and compared the performance of a computer vision application for counting vehicles in a parking lot across fog, mist, edge and cloud scenarios, along with a Monte Carlo simulation of parking occupancy using a natural language interface.
 - **Spot-Wise Smart Parking: An Edge-Enabled Architecture with YOLOv11 and Digital Twin Integration**: paper currently under review by SBC (2025).
 - **Image-Based Time Series Forecasting**. Next-Day Electricity Demand Forecasting using Computer Vision and Deep Neural Networks. (2022-2023).
 - **San Marcos Smart City: A proposed framework for developing Smart City services based on the ISO 37120:2018 standard for Lima**. (2020-2021)
 - **Planning study for the development of the FIEE UNI Smart Library**. (2020-2021)

- **Development of a Smart Campus FIEE system and web application for real-time visualization of the UNI Bus location using LoRaWAN IoT technologies.** (2020-2021)
- **Prototype for hand-to-face movement recognition to fight COVID-19 for a vulnerable population using an IMU sensor and a deep learning model embedded in an SBC.** (2020)
- **Mobile COVID-19 app based on RSSI signals to enforce social distancing using smartphones' Bluetooth signal.** (2020)
- **Smart campus wayfinding system for visitors based on Bayesian filters.** (2019)

Technologies and Languages

- Languages: Python, JavaScript, C++, bash script.
- Technologies: FIWARE, Docker, React, AWS, GCP, TensorFlow, Edge Impulse, Mosquitto, Node-RED.
- Other: LoRaWAN, Sigfox, MQTT, Arduino, ESP32, Raspberry, Solidworks, AutoCAD.

Projects

- **Internet of Things Technology Training for Professors of the Academic Department of Systems and Computer Engineering at UNAMAD University.** IoT workshop course: LoRaWAN, Cloud/Edge Computing, ML forecasting and web application on AWS (2023).
- **Course "Research methodology leading to the development of research projects in Systems and Computer Engineering".** IoT course prepared for UNAMAD University. Modules: LoRaWAN, Cloud/Edge Computing, TinyML, Machine Learning and web application development for IoT projects (2022).
- **Design, construction and parameter monitoring of an NFT hydroponic module for growing lettuce.** Design, measurement and construction of an NFT (Nutrient Film Technique) hydroponic module for real-time information monitoring (2022-2023). Project paused since starting the master's program.
- **Design of a management and tracking platform for a transportation company.** application to automate the registration of Reference Guides with React and AWS Amplify to reduce manual entry time, plus a visualization dashboard built with D3.js (2022).
- **Integration of an Industrial OPC Server and Variable Monitoring on AWS.** Data extraction from an Industrial OPC Classic Server and integration with AWS IoT Core for real-time data visualization on an Elasticsearch dashboard (2021).
- **Training at Universidad Nacional Amazónica Madre de Dios (UNAMAD).** IoT workshops using LoRaWAN and AWS technologies (2021).

Awards and Distinctions

- **Ñahui Health.** IoT system for real-time monitoring and alerting of oxygen tank pressure, using computer vision with an ESP32-CAM on the AWS cloud and notifications via Telegram and email. Fifth place: Bicentennial Congress Technology Works and Projects Competition "Peruvian Engineering 200 Years of the Republic" (2021)
- **Innovation Distinction at the 2021 National Digital Democracy Award** awarded to IoT Community Peru. (2021)
- **Applied Technology Development Award at the IEEE INTERCON 2019 technology fair.** Project "Real-time IoT-based monitoring system for efficient management of irrigation service and park maintenance using Sigfox and ML technologies". (2019)
- **Winner of the Hackster Bounty 'Alexa & Arduino Smart Home Challenge'.** Contest sponsored by Amazon and Arduino. Project "Smart Shower Gadget". (2018)

Interests

- I enjoy taking part in marathons, swimming, gymnastics, writing stories, oil painting, and playing the guitar.